

QuantaPlex Series

T41S-2U/T41SP-2U

**2U 4-Node Server Featuring
Latest DDR4 Technology**

User's Guide

The Quanta logo is rendered in a bold, italicized, blue sans-serif font. The letters are closely spaced, and the overall style is modern and professional.

Version: 2.0.0

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All the illustrations in this technical guide are for reference only and are subject to change without prior notice.

Intended Application Uses

This product was evaluated as Information Technology Equipment (ITE), which may be installed in offices, schools, computer rooms, and similar commercial type locations. The suitability of this product for other product categories and environments (such as medical, industrial, residential, alarm systems, and test equipment), other than an ITE application, may require further evaluation.

Conventions

Several different typographic conventions are used throughout this manual. Refer to the following examples for common usage.

Bold type face denotes menu items, buttons and application names.

Italic type face denotes references to other sections, and the names of the folders, menus, programs, and files.

<**Enter**> type face denotes keyboard keys.



WARNING!

Warning information appears before the text it references and should not be ignored as the content may prevent damage to the device.



CAUTION!

CAUTIONS APPEAR BEFORE THE TEXT IT REFERENCES, SIMILAR TO NOTES AND WARNINGS. CAUTIONS, HOWEVER, APPEAR IN CAPITAL LETTERS AND CONTAIN VITAL HEALTH AND SAFETY INFORMATION.

Note:

Highlights general or useful information and tips.

About the System

Chapter 1

This section introduces the system, its different configuration(s) and the main features.

1.1 Introduction

This manual is written for system technicians who are responsible for troubleshooting, upgrading, and repairing the server chassis. This document provides an overview of the hardware features of the chassis, troubleshooting information, and instructions on how to add and replace components of the multi-node server series. The document also provides information on the BIOS, and Baseboard Management Controller (BMC).

For the latest version of this manual, see www.QuantaQCT.com.

System Features

The system comprises a 2U/30.5" long chassis using a standard SSI mainboard. Major features per sled include:

- **Chipset:** Intel® C610 series
- **Processors (x2):** Intel® Xeon E5-2600 v3 series processor
- **PCIe slots:**
 - (1) PCIe x16 Gen3 riser slot for low-profile card per node
 - (1) PCIe x8 Gen3 mezzanine slot for Quanta mezzanine card per node
 - (1) PCIe x8 Gen3 mezzanine slot for Quanta OCP mezzanine card per node
- **Memory:** Up to sixteen DIMM slots are available. DDR4 2133 MT/s RDIMM, and LRDIMM memory are supported.
- **Network*:** (1) Quanta 1GbE dual port OCP mezzanine card (Optional), (1) Quanta 10GbE SFP+ OCP dual port mezzanine card (Optional), (1) Quanta 10GbE dual port OCP mezzanine card (Optional)

*Visit QuantaQCT for the latest Network support listings.

Note:

The system supports: 1600W 200V-240V and 1200W 100V-240V.

Table 1: 2U4N Specification

SPECIFICATION	DESCRIPTION
Form factor (W x H x D)	• X5M (2U chassis) • 444mm x 87.5mm x 790mm / 17.48" x 3.44" x 31.1"
Mainboard form factor (W x L), half-width	• 165mm x 508mm / 6.5" x 20" • Up to 4 independent mainboard sleds • 2.5" SKU - 6 x 2.5" HDD per node • 3.5" SKU - 3 x 3.5" HDD per node
Processor	(2) Intel® Xeon® processors E5-2600 v3 family per node, up to 145W

Table 1: 2U4N Specification (Continued)

SPECIFICATION	DESCRIPTION
Max. processor wattage	145W, optimized power delivery for 95W, VRD 12.5
QPI speed	9.6 GT/s, 8.0GT/s, 6.4GT/s
Chipset	Intel® C610 (Wellsburg)
Memory	(16) DDR4 2133 MT/s ECC RDIMM/LRDIMM slots per node, up to 512 GB
Storage	• (3) 3.5" hot-plug or • (6) 2.5" hot-plug HDD per node
HDD backplane	1 to 1
PCIe Expansion slot	• (1) PCIe x16 Gen3 riser slot for low-profile card per node • (1) PCIe x8 Gen3 Mezzanine slot for Quanta Mezzanine card per node • (1) PCIe x8 Gen3 OCP Mezzanine slot for OCP card per node
HW RAID options	• Quanta LSI® 2108 6Gb/s RAID mezzanine, RAID 0, 1, 5, 10. RAID 6 with additional RAID key • Quanta LSI® 2208 6Gb/s RAID mezzanine, RAID 0, 1, 5, 10. RAID 6 with additional RAID key
Network options	Quanta OCP Mezzanine • (1) Quanta 1Gb Ethernet dual port OCP mezzanine card (Optional) • (1) Quanta 10Gb SFP+ dual port OCP mezzanine card (Optional) • (1) Quanta 10Gb Ethernet dual port OCP mezzanine card (Optional) Quanta Mezzanine • (1) Quanta 1Gb Ethernet dual port mezzanine card • (1) Quanta 10Gb Ethernet dual port mezzanine card • (1) Quanta 40Gb Ethernet mezzanine card • (1) Quanta 40Gb IB QDR mezzanine card • (1) Quanta 56Gb IB FDR mezzanine card
Management port	(1) Dedicated 10/100 BASE-T RJ45 management port per node
Integrated graphics BMC	ASPEED AST2400, Ethernet SW LAN9303 to BMC for dedicated management NIC port.
Front I/O	(1) USB 2.0 port per node
Rear I/O	• (2) USB 3.0 ports per node • (1) VGA per node • (1) RS232 Serial port per node • (1) 10/100 BASE-T RJ45 management port per node
Power supply unit	• (2) 1600W high efficiency redundant PSUs, 200-240Vac, 8A, 50/60Hz • (2) 1200W high efficiency redundant PSUs, 100-240Vac, 8A-12A, 50/60Hz
Rating (per PSU inlet)	(1) 200-240Vac, 8A, 50/60Hz; (2) 100-240Vac 10A-6.7A; 50/60Hz
RoHS	Yes
Intel node management support	Yes

Table 1: 2U4N Specification (Continued)

SPECIFICATION	DESCRIPTION
System management	IPMI v2.0 Compliant, on board "KVM over IP" support
Operating environment	- Operating temperature: 5°C to 35°C (41°F to 95°F) - Non-operating temperature: -40°C to 70°C (-40°F to 158°F) - Operating relative humidity: 50% to 85%RH. - Non-operating relative humidity: 20% to 90%RH

Product Configuration Limitation

Table 2: Product Configuration Limitation

ITEM		3.5"		2.5"	
CPU		135W (1U cooling)	145W or 2U cooling	135W (1U cooling)	145W or 2U cooling
Heatsink	CPU0	Standard	Vapor Chamber HS	Standard	Vapor Chamber HS
	CPU1	Standard	Vapor Chamber HS	Standard	Vapor Chamber HS
Configuration	Storage	12 x HDD	8 x HDD	24 x HDD	12 x HDD
	DIMM	16 x DIMM	12 x DIMM	16 x DIMM	12 x DIMM
N+1 Fan rotor redundancy		Ambient 30C	Ambient 30C	Ambient 30C	Ambient 25C
Quanta 10GbE RJ45 OCP mezzanine		8x HDD if CPU is over 105W		12x HDD if CPU is over 105W	

1.2 A Tour of the System

System Overview

The system is available in two models: 24 x 2.5" HDD or 12 x 3.5" HDD model. The following illustrations describe the major components of these two variants.

2.5" HDD System

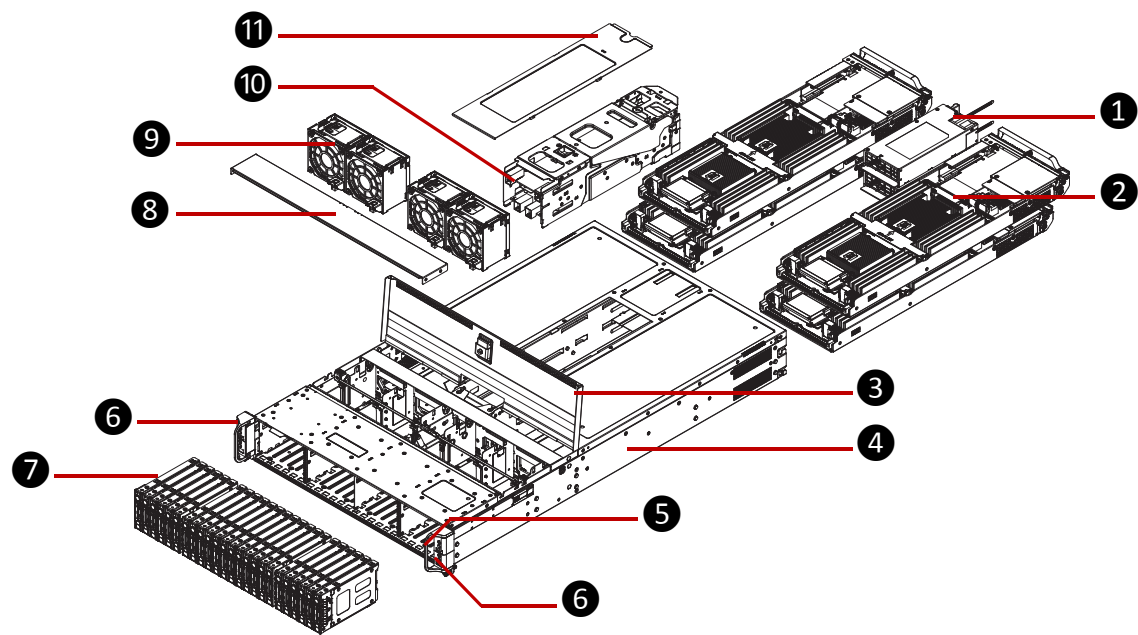


Figure 1-1. 2.5" HDD System Component Overview

Table 3: 2.5" System Component Overview

No.	ITEM	DESCRIPTION
1	Power supply unit (PSU)	Two 1600W high efficiency redundant PSU, 200-240Vac, 8A, 50/60Hz
		Two 1200W high efficiency redundant PSU, 100-240Vac, 8A-12A, 50/60Hz
2	MB sled	Hot-swap mainboard sleds (x4)
3	Fan assembly cover	Open to service fan modules
4	Chassis	2U System chassis 444mm/17.48"(W)x 87.5mm/3.44"(H)x 790mm/31.1"(D)
5	Server handle	Two server handles used for pulling the system out of the rack CAUTION! THE HANDLES ARE DESIGNED FOR THE EXTENSION OF THE SYSTEM FROM THE RACK. THE HANDLES ARE NOT DESIGNED TO CARRY THE WEIGHT OF THE SYSTEM. DO NOT USE THE HANDLES TO MOVE OR LIFT THE SYSTEM.
6	Front control panels	Front control panels 1/2 (left) and 3/4 (right). See <i>Front Control Panel and Status LED</i> on page 1-10 for further information

Table 3: 2.5" System Component Overview (Continued)

No.	ITEM	DESCRIPTION
7	2.5" HDD bay	2.5" hard disk drive (HDD) modules (x24)
8	Neck-top cover	Remove to service HDD backplane. Only available in 2.5" HDD model.
9	Fan modules	Four 80 mm (3.14") x 56 mm (2.20") variable-speed fans
10	PDB assembly	Two power distribution boards (2U4N) and single cage
11	Power distribution board assembly cover	Remove to service the PDB assembly

3.5" HDD System

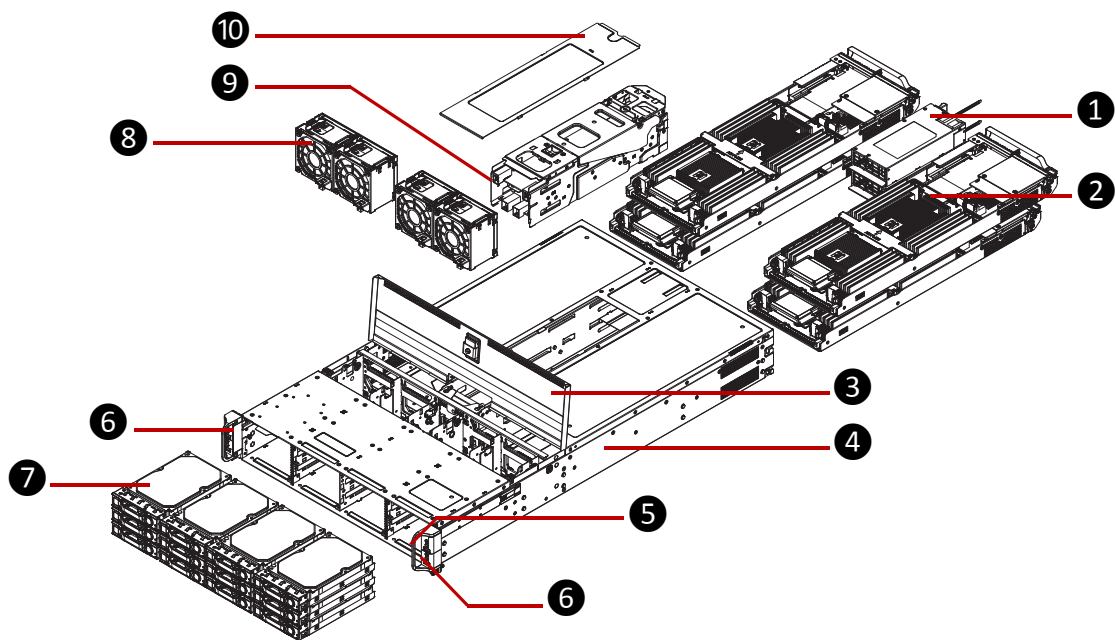


Figure 1-2. 3.5" System Component Overview

Table 4: 3.5" System Component Overview

No.	ITEM	DESCRIPTION
1	Power supply unit	Two 1600W high efficiency redundant PSU, 200-240Vac, 8A, 50/60Hz
		Two 1200W high efficiency redundant PSU, 100-240Vac, 8A-12A, 50/60Hz
2	MB sled	Hot-swap mainboard sleds (x4)
3	Fan assembly cover	Open to service fan modules
4	Chassis	2U System chassis 444mm/17.48"(W)x 87.5mm/3.44"(H)x 790mm/31.1"(D)

Table 4: 3.5" System Component Overview (Continued)

No.	ITEM	DESCRIPTION
5	Server handle	Two server handles used for pulling the system out of the rack CAUTION! THE HANDLES ARE DESIGNED FOR THE EXTENSION OF THE SYSTEM FROM THE RACK. THE HANDLES ARE NOT DESIGNED TO CARRY THE WEIGHT OF THE SYSTEM. DO NOT USE THE HANDLES TO MOVE OR LIFT THE SYSTEM.
6	Front control panels	Front control panels 1/2 (left) and 3/4 (right). See <i>Front Control Panel and Status LED</i> on page 1-10 for further information
7	3.5" HDD bay	3.5" hard disk drive (HDD) modules (x12)
8	Fan modules	Four 80 mm (3.14") x 56 mm (2.20") variable-speed fans
9	PDB assembly	Two power distribution boards (2U4N) and single cage
10	Power distribution board cover	Remove to service the PDB assembly

System Front View

2.5" HDD SKU

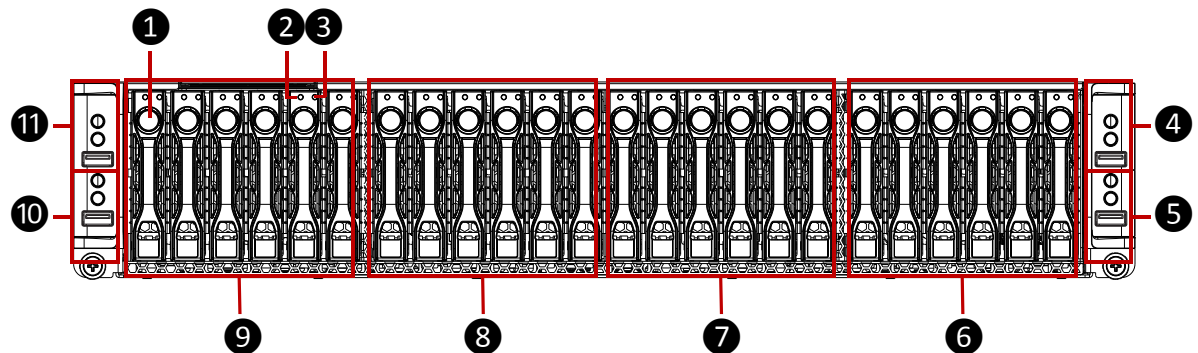


Figure 1-3. Front View for 2.5" HDD Configuration

Table 5: System Front View for 2.5" HDD Configuration

No	FEATURE	DESCRIPTION
1	HDD release lever & button	Button and release lever for removing HDD tray
2	HDD locate LED	Blue: Healthy Amber: Warning
3	HDD active LED	Blue: Access
4	Front control panel (Node 3)	Includes power/ID button, power/ID/status LED, and USB port
5	Front control panel (Node 4)	

Table 5: System Front View for 2.5" HDD Configuration (Continued)

No	FEATURE	DESCRIPTION
6	Node 4	MB4 HDD Bay
7	Node 3	MB3 HDD Bay
8	Node 2	MB2 HDD Bay
9	Node 1	MB1 HDD Bay
10	Front control panel (Node 2)	Includes power/ID button, power/ID/status LED, and USB port
11	Front control panel (Node 1)	

3.5" HDD SKU

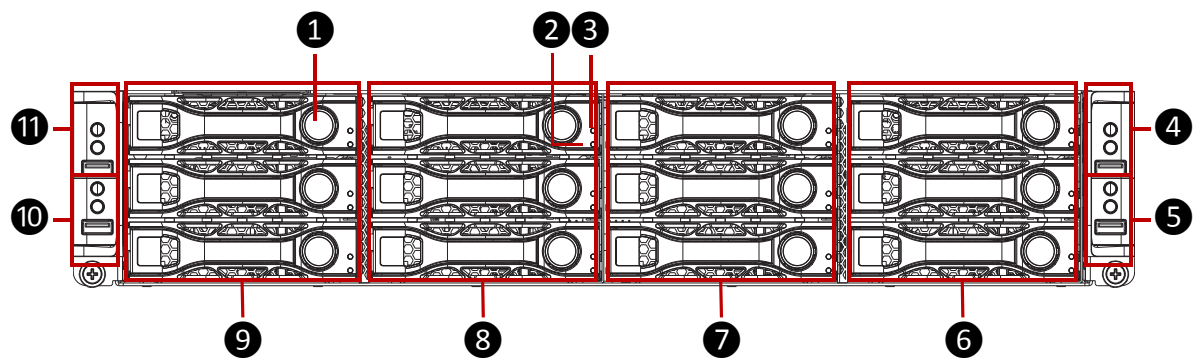


Figure 1-4. Front View for 3.5" HDD Configuration

Table 6: System Front View for 3.5" HDD Configuration

No.	FEATURE	DESCRIPTION
1	HDD release lever & button	Button and release lever for removing HDD tray
2	HDD active LED	Blue: Access
3	HDD locate LED	Blue: Healthy Amber: Warning
4	Front control panel (Node 3)	Includes power/ID button, power/ID/status LED, and USB port
5	Front control panel (Node 4)	
6	Node 4	MB4 HDD Bay
7	Node 3	MB3 HDD Bay
8	Node 2	MB2 HDD Bay
9	Node 1	MB1 HDD Bay

Table 6: System Front View for 3.5" HDD Configuration (Continued)

No.	FEATURE	DESCRIPTION
10	Front control panel (Node 2)	Includes power/ID button, power/ID/status LED, and USB port
11	Front control panel (Node 1)	

Front Control Panel and Status LED

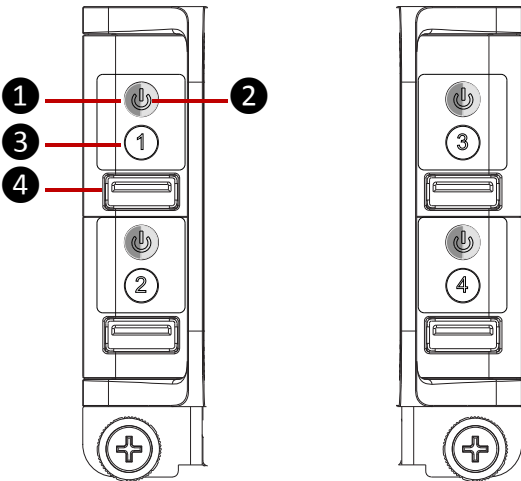


Figure 1-5. Front Control Panel

Table 7: Front Control Panel

No	FEATURE	DESCRIPTION
1	Power button with LED	Blue on: System power on Off: System power off Amber blinking: DC off and fault Amber and blue blinking: DC on and fault
2	Status LED	Amber blinking: System failure
3	ID button with LED	Blue blinking: ID indicator Off: No indicator 1: MB1 (2: MB2, 3: MB3, 4: MB4)
4	USB 2.0 port	USB 2.0 port

Rear View

The system rear consists of four mainboard sleds and two power supply units. Dummy modules are installed to maintain proper airflow in the event that mainboard slots are not fully populated.

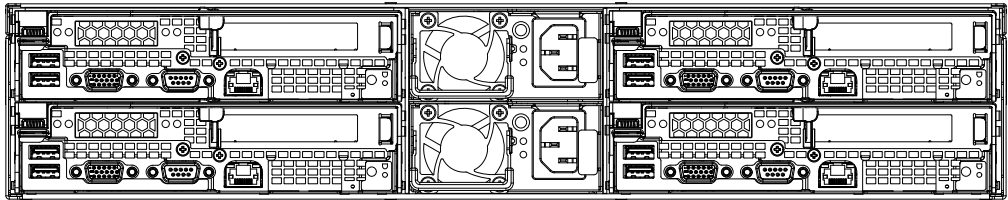


Figure 1-6. System Rear View

Rear I/O Ports

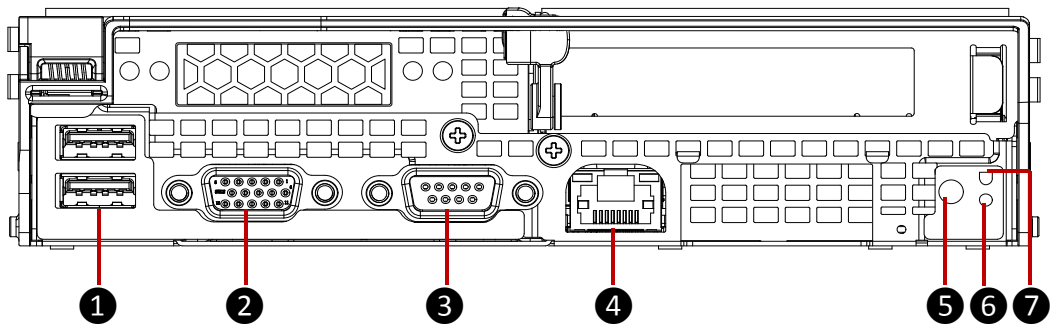


Figure 1-7. Rear I/O Ports

Table 8: Rear I/O Ports

No	FEATURE	DESCRIPTION
1	USB ports	Dual 3.0 USB ports. Connect to USB device.
2	VGA port	Standard VGA compatible, 15-pin connector supporting up to 1920 x 1200 32bpp@60Hz resolution. Connect to display device.
3	Serial port	Standard DB9 serial port. Connect to serial device.
4	Management port	Dedicated BMC LAN port (RJ45) for remote control / management.
5	Power button	Press and hold the power button more than four seconds to turn off the motherboard.
6	Mainboard status LED	Off: normal Amber (blinking): BMC reports system health fault
7	Power LED	Off: power off Blue (blinking): BMC reports system health fault

Note:

The USB connector must be:
No bigger than 17mm/ 0.67" (W) x 8 mm / 0.31" (H) to avoid interference with the other ports.

Note:

The temperature through the ventilation holes was high. Make sure to use a cable rated to withstand 75 degrees or higher.

HDD Configuration

The multi-node systems are available as a 2.5" or 3.5" SKU. Within the HDD array, the HDD enumeration is as follows:

2.5" HDD Configuration

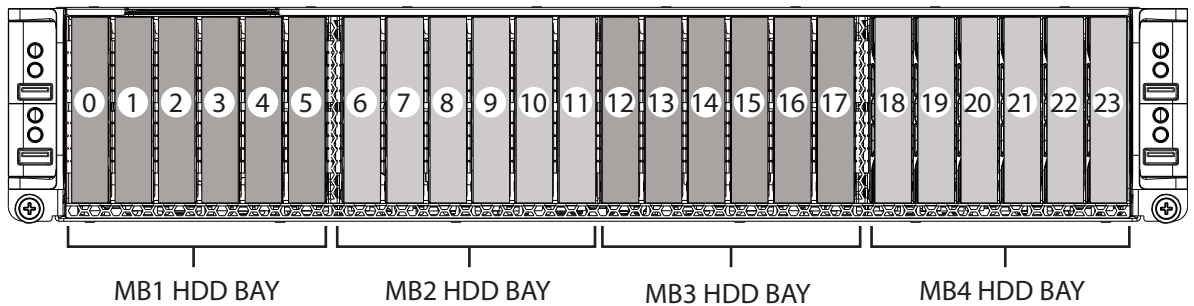


Figure 1-8. 2.5" HDD Configuration

3.5" HDD Configuration

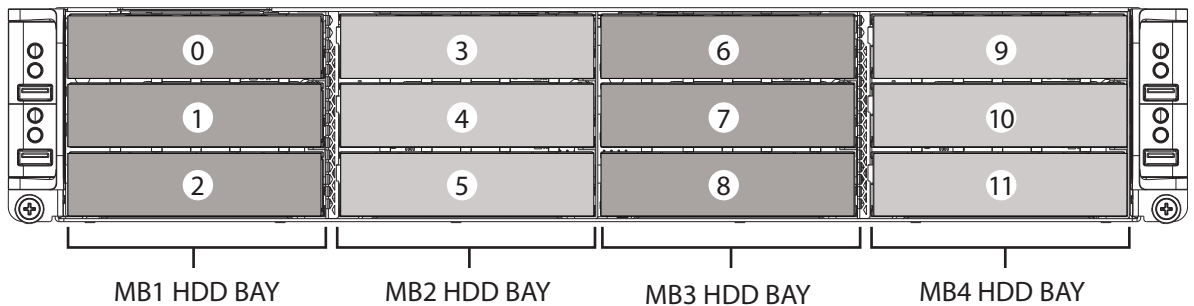


Figure 1-9. 3.5" HDD Configuration

LAN LEDs

The system with Quanta OCP mezzanine supports two LAN ports. See the following illustration and table for details.

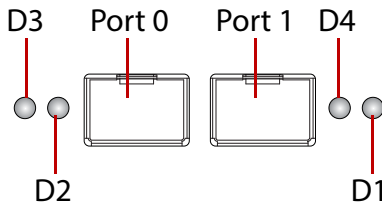


Figure 1-10. LAN Mezzanine Card

Table 9: LAN LED Behavior

LED		COLOR	DISPLAY STATUS	DESCRIPTION
Port 0	Link / Activity (D2)	Green	ON	LAN link
		Green	Blink	LAN access (off when there is traffic)
		-	OFF	Disconnect
	Speed (D3)	Green	ON	Green, link speed is 10 Gbits/sec
		Amber	ON	Amber, link speed is 1 Gbits/sec
		-	OFF	OFF, link speed is 100 Mbits/sec
Port 1	Link / Activity (D1)	Green	ON	LAN link
		Green	Blink	LAN access (off when there is traffic)
		-	OFF	Disconnect
	Speed (D4)	Green	ON	Green, link speed is 10 Gbits/sec
		Amber	ON	Amber, link speed is 1 Gbits/sec
		-	OFF	OFF, link speed is 100 Mbits/sec

Power Sub-System

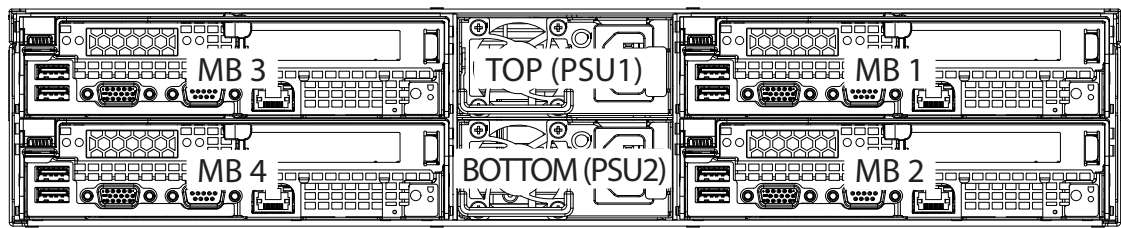


Figure 1-11. PSU to Mainboard Sled Description



WARNING!

Mainboard combinations other than the system are not supported.

A system has two modular Power Supply Units (PSU). Both PSUs are directly connected to the Power Distribution Boards (PDBs), the HDD backplane, and middle plane allowing each PSU to individually provide power for all mainboards (MB1 to MB4).

The power supply units supported: 2 x 1600W redundant PSU @ 200V-240Vac, 8A, 50/60Hz or 2 x 1200W redundant PSU @ 100V-240Vac, 8A-12A, 50/60Hz.

PSU LED

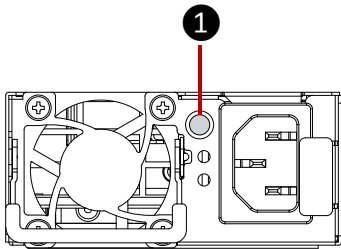


Figure 1-12. PSU LED

Table 10: PSU LED Description

No	FEATURE	STATUS	DESCRIPTION
1	PSU LED	Green	Normal operation
		Amber: blinking	Power off, fault
		Green: blinking	Power on, standby mode

Cooling Sub-System

Fans may spin for some time after the system has been powered off. Allow time for the fans to stop rotating before handling system components.

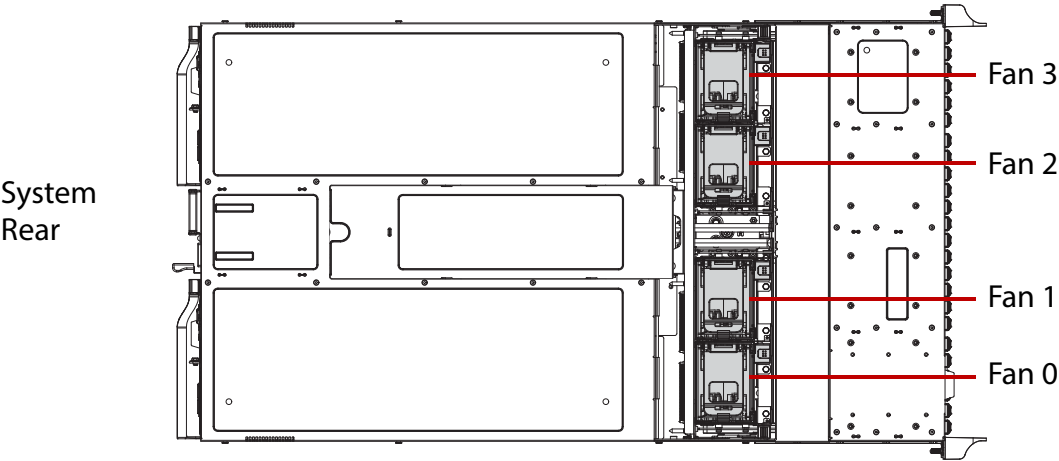


Figure 1-13. Cooling Sub System (2.5" HDD model shown)

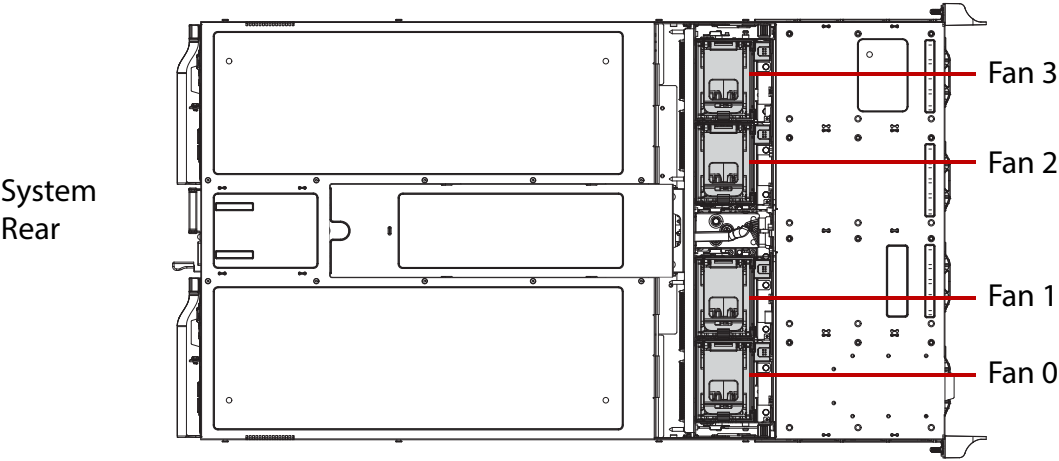


Figure 1-14. Cooling Sub System (3.5" HDD model shown)

Regulatory and Compliance Information

Chapter 2

This section provides regulatory and compliance information applicable to this system.

Server Safety Information

To reduce the risk of bodily injury, electrical shock, fire, and equipment damage, read this document and observe all warnings and precautions in this guide before installing or maintaining your server product.

In the event of a conflict between the information in this document and information provided with the product or on the website for a particular product, the product documentation takes precedence.

Your server should be integrated and serviced only by technically qualified persons.

You must adhere to the guidelines in this guide and the assembly instructions in your server manuals to ensure and maintain compliance with existing product certifications and approvals. Use only the described, regulated components specified in this guide. Use of other products / components will void the UL Listing and other regulatory approvals of the product, and may result in noncompliance with product regulations in the region(s) in which the product is sold.

Safety Warnings and Cautions

To avoid personal injury or property damage, before you begin installing the product, read, observe, and adhere to all of the following safety instructions and information. The following safety symbols may be used throughout the documentation and may be marked on the product and / or the product packaging.

CAUTION	Indicates the presence of a hazard that may cause minor personal injury or property damage if the CAUTION is ignored.
WARNING	Indicates the presence of a hazard that may result in serious personal injury if the WARNING is ignored.
	Indicates potential hazard if indicated information is ignored.
	Indicates shock hazards that result in serious injury or death if safety instructions are not followed.
	Indicates hot components or surfaces.
	Indicates do not touch fan blades, may result in injury.
	Indicates to unplug all AC power cord(s) to disconnect AC power.
	Please recycle battery.
	The rail racks are designed to carry only the weight of the server system. Do not use rail-mounted equipment as a workspace. Do not place additional load onto any rail-mounted equipment.
	Indicates two people are required to safely handle the system.

Intended Application Uses

This product was evaluated as Information Technology Equipment (ITE), which may be installed in offices, schools, computer rooms, and similar commercial type locations. The suitability of this product for other product categories and environments (such as medical, industrial, residential, alarm systems, and test equipment), other than an ITE application, may require further evaluation.

Site Selection

The system is designed to operate in a typical office environment. Choose a site that is:

- Clean, dry, and free of airborne particles (other than normal room dust).
- Well-ventilated and away from sources of heat including direct sunlight and radiators.
- Away from sources of vibration or physical shock.
- Isolated from strong electromagnetic fields produced by electrical devices.
- In regions that are susceptible to electrical storms, we recommend you plug your system into a surge suppressor and disconnect telecommunication lines to your modem during an electrical storm.
- Provided with a properly grounded wall outlet.
- Provided with sufficient space to access the power supply cord(s), because they serve as the product's main power disconnect.
- Provided with either two independent AC power sources or two independent phases from a single source.

Equipment Handling Practices

Reduce the risk of personal injury or equipment damage:

- Conform to local occupational health and safety requirements when moving and lifting equipment.
- Use mechanical assistance or other suitable assistance when moving and lifting equipment.
- To reduce the weight for easier handling, remove any easily detachable components.

Power and Electrical Warnings

Caution: The power button, indicated by the stand-by power marking, DOES NOT completely turn off the system AC power, 5V standby power is active whenever the system is plugged in. To remove power from system, you must unplug the AC power cord from the wall outlet. Your system may use more than one AC power cord. Make sure all AC power cords are unplugged. Make sure the AC power cord(s) is / are unplugged before you open the chassis, or add or remove any non hot-plug components.

Do not attempt to modify or use an AC power cord if it is not the exact type required. A separate AC cord is required for each system power supply.

Some power supplies in servers use Neutral Pole Fusing. To avoid risk of shock use caution when working with power supplies that use Neutral Pole Fusing.

The power supply in this product contains no user-serviceable parts. Do not open the power supply. Hazardous voltage, current and energy levels are present inside the power supply. Return to manufacturer for servicing.

When replacing a hot-plug power supply, unplug the power cord to the power supply being replaced before removing it from the server.

To avoid risk of electric shock, turn off the server and disconnect the power cord, telecommunications systems, networks, and modems attached to the server before opening it.

Power Cord Warnings

If an AC power cord was not provided with your product, purchase one that is approved for use in your country.

Caution: To avoid electrical shock or fire, check the power cord(s) that will be used with the product as follows:

- Do not attempt to modify or use the AC power cord(s) if they are not the exact type required to fit into the grounded electrical outlets.
- The power cord(s) must meet the following criteria:
 - The power cord must have an electrical rating that is greater than that of the electrical current rating marked on the product.
 - The power cord must have safety ground pin or contact that is suitable for the electrical outlet.
 - The power supply cord(s) is / are the main disconnect device to AC power. The socket outlet(s) must be near the equipment and readily accessible for disconnection.
 - The power supply cord(s) must be plugged into socket-outlet(s) that is /are provided with a suitable earth ground.

System Access Warnings

Caution: To avoid personal injury or property damage, the following safety instructions apply whenever accessing the inside of the product:

- Turn off all peripheral devices connected to this product.
- Turn off the system by pressing the power button to off.
- Disconnect the AC power by unplugging all AC power cords from the system or wall outlet.
- Disconnect all cables and telecommunication lines that are connected to the system.
- Retain all screws or other fasteners when removing access cover(s). Upon completion of accessing inside the product, refasten access cover with original screws or fasteners.
- Do not access the inside of the power supply. There are no serviceable parts in the power supply. Return to manufacturer for servicing.
- Power down the server and disconnect all power cords before adding or replacing any non hot-plug component.
- When replacing a hot-plug power supply, unplug the power cord to the power supply being replaced before removing the power supply from the server.

Caution: If the server has been running, any installed processor(s) and heat sink(s) may be hot.

Unless you are adding or removing a hot-plug component, allow the system to cool before opening the covers. To avoid the possibility of coming into contact with hot component(s) during a hot-plug installation, be careful when removing or installing the hot-plug component(s).

Caution: To avoid injury do not contact moving fan blades. If your system is supplied with a guard over the fan, do not operate the system without the fan guard in place.

Rack Mount Warnings

Note: *The following installation guidelines are required by UL for maintaining safety compliance when installing your system into a rack.*

The equipment rack must be anchored to an unmovable support to prevent it from tipping when a server or piece of equipment is extended from it. The equipment rack must be installed according to the rack manufacturer's instructions.

Install equipment in the rack from the bottom up, with the heaviest equipment at the bottom of the rack.

Extend only one piece of equipment from the rack at a time.

You are responsible for installing a main power disconnect for the entire rack unit. This main disconnect must be readily accessible, and it must be labeled as controlling power to the entire unit, not just to the server(s).

To avoid risk of potential electric shock, a proper safety ground must be implemented for the rack and each piece of equipment installed in it.

Elevated Operating Ambient - If installed in a closed or multi-unit rack assembly, the operating ambient temperature of the rack environment may be greater than room ambient. Therefore, consideration should be given to installing the equipment in an environment compatible with the maximum ambient temperature (T_{ma}) specified by the manufacturer.

Reduced Air Flow - Installation of the equipment in a rack should be such that the amount of air flow required for safe operation of the equipment is not compromised.

Mechanical Loading - Mounting of the equipment in the rack should be such that a hazardous condition is not achieved due to uneven mechanical loading.

Circuit Overloading - Consideration should be given to the connection of the equipment to the supply circuit and the effect that overloading of the circuits might have on overcurrent protection and supply wiring. Appropriate consideration of equipment nameplate ratings should be used when addressing this concern.

Reliable Earthing - Reliable earthing of rack-mounted equipment should be maintained.

Particular attention should be given to supply connections other than direct connections to the branch circuit (e.g. use of power strips).

Electrostatic Discharge (ESD)

Caution: *ESD can damage drives, boards, and other parts. We recommend that you perform all procedures at an ESD workstation. If one is not available, provide some ESD protection by wearing an antistatic wrist strap attached to chassis ground -- any unpainted metal surface -- on your server when handling parts.*

Always handle boards carefully. They can be extremely sensitive to ESD. Hold boards only by their edges. After removing a board from its protective wrapper or from the server, place the board component side up on a grounded, static free surface. Use a conductive foam pad if available but not the board wrapper. Do not slide board over any surface.

Other Hazards

Battery Replacement

Caution: *There is the danger of explosion if the battery is incorrectly replaced. When replacing the battery, use only the battery recommended by the equipment manufacturer.*

Dispose of batteries according to local ordinances and regulations.

Do not attempt to recharge a battery.

Do not attempt to disassemble, puncture, or otherwise damage a battery.

Cooling and Airflow

Caution: Carefully route cables as directed to minimize airflow blockage and cooling problems. For proper cooling and airflow, operate the system only with the chassis covers installed. Operating the system without the covers in place can damage system parts. To install the covers:

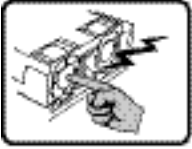
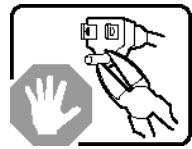
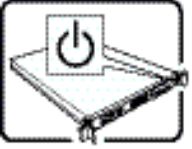
- Check first to make sure you have not left loose tools or parts inside the system.
- Check that cables, add-in cards, and other components are properly installed.
- Attach the covers to the chassis according to the product instructions.

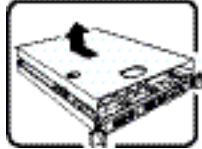
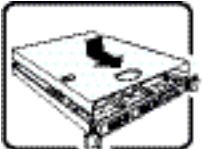
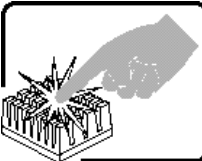
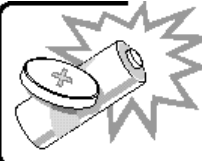
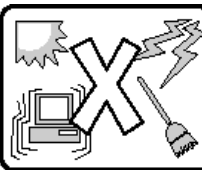
Laser Peripherals or Devices

Caution: To avoid risk of radiation exposure and / or personal injury:

- Do not open the enclosure of any laser peripheral or device
- Laser peripherals or devices have are not serviceable
- Return to manufacturer for servicing

Use certified Optical Fiber Transceiver Class I Laser Product

	The power supply in this product contains no user-serviceable parts. Refer servicing only to qualified personnel.
	Do not attempt to modify or use the supplied AC power cord if it is not the exact type required. A product with more than one power supply will have a separate AC power cord for each supply.
	The power button on the system does not turn off system AC power. To remove AC power from the system, you must unplug each AC power cord from the wall outlet or power supply. The power cord(s) is considered the disconnect device to the main (AC) power. The socket outlet that the system plugs into shall be installed near the equipment and shall be easily accessible.
	SAFETY STEPS: Whenever you remove the chassis covers to access the inside of the system, follow these steps: 1. Turn off all peripheral devices connected to the system. 2. Turn off the system by pressing the power button. 3. Unplug all AC power cords from the system or from wall outlets. 4. Label and disconnect all cables connected to I/O connectors or ports on the back of the system. 5. Provide some electrostatic discharge (ESD) protection by wearing an antistatic wrist strap attached to chassis ground of the system-any unpainted metal surface-when handling components. 6. Do not operate the system with the chassis covers removed.

	After you have completed the six SAFETY steps above, you can remove the system covers. To do this: 1. Unlock and remove the padlock from the back of the system if a padlock has been installed. 2. Remove and save all screws from the covers. 3. Remove the cover(s).
	For proper cooling and airflow, always reinstall the chassis covers before turning on the system. Operating the system without the covers in place can damage system parts. To install the covers: 1. Check first to make sure you have not left loose tools or parts inside the system. 2. Check that cables, add-in cards, and other components are properly installed. 3. Attach the covers to the chassis with the screws removed earlier, and tighten them firmly. 4. Insert and lock the padlock to the system to prevent unauthorized access inside the system. 5. Connect all external cables and the AC power cord(s) to the system.
	A microprocessor and heat sink may be hot if the system has been running. Also, there may be sharp pins and edges on some board and chassis parts. Contact should be made with care. Consider wearing protective gloves.
	Danger of explosion if the battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the equipment manufacturer. Dispose of used batteries according to manufacturer's instructions.
	The system is designed to operate in a typical office environment. Choose a site that is: • Clean and free of airborne particles (other than normal room dust). • Well ventilated and away from sources of heat including direct sunlight. • Away from sources of vibration or physical shock. • Isolated from strong electromagnetic fields produced by electrical devices. • In regions that are susceptible to electrical storms, we recommend you plug your system into a surge suppressor and disconnect telecommunication lines to your modem during an electrical storm. • Provided with a properly grounded wall outlet. • Provided with sufficient space to access the power supply cord(s), because they serve as the product's main power disconnect.
	The rail racks are designed to carry only the weight of the server system. Do not use rail-mounted equipment as a workspace. Do not place additional load onto any rail-mounted equipment.



Heavy object. Indicates two people are required to safely handle the system.

Product Regulatory Compliance Markings

This product is marked with the following Product Certification Markings:

Product Regulatory Compliance Markings

Regulatory Compliance	Region	Marking
cULus Listing Mark	USA / Canada	
CE Mark	Europe	
FCC Marking (Class A)	USA	This device complies with Part 15 of the FCC Rules. Operation of this device is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) This device must accept any interference received, including interference that may cause undesired operation.
ICES	Canada	CAN ICES-3 (A)/NMB-3(A)
VCCI Marking (Class A)	Japan	この装置は、クラス A 情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。 VCCI-A
BSMI Certification Number & Class A Warning	Taiwan	 警告使用者： 此為甲類資訊技術設備，於居住環境使用中時，可能會造成射頻擾動，在此種情況下，使用者會被要求採取某些適當的對策。
EAC Marking	Russia	
CCC	China	 此为 A 级产品,在生活环境,该产品可能会造成无线电干扰。在这种情况下,可能需要用户对其干扰采取切实可行的措施。
Recycling Package Mark	Other than China	
MSIP	Korea	 A급 기기 (업무용 정보통신기기) 이 기기는 업무용으로 전자파적합등록을 한 기기이오니 판매자 또는 사용자는 이 점을 주의하시기 바라며, 만약 잘못판매 또는 구입하였을 때에는 가정용으로 교환하시기 바랍니다.

Electromagnetic Compatibility Notices

FCC Verification Statement (USA)

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Europe (CE Declaration of Conformity)

This product has been tested in accordance too, and complies with the Low voltage Directive(2006/95/EC) and EMC Directive (2004/108/EC). The product has been marked with the CE Mark to illustrate its compliance.

VCCI (Japan)

この装置は、クラス A 情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。

VCCI-A

English translation of the notice above:

This is a Class A product based on the standard of the Voluntary Control Council for Interference (VCCI) from Information Technology Equipment. If this is used near a radio or television receiver in a domestic environment, it may cause radio interference. Install and use the equipment according to the instruction manual.

BSMI (Taiwan)

The BSMI Certification Marking and EMC warning is located on the outside rear area of the product.

警告使用者：

此為甲類資訊技術設備，於居住環境使用中時，可能會造成射頻擾動，在此種情況下，使用者會被要求採取某些適當的對策。

MSIP (Korea)

Ministry of Science, ICT & Future Planning (MSIP) Class A Statement:

A급 기기 (업무용 정보통신기기)

이 기기는 업무용으로 전자파적합등록을 한 기기이오니 판매자 또는 사용자는 이 점을 주의하시기 바라며, 만약 잘못판매 또는 구입하였을 때에는 가정용으로 교환하시기 바랍니다.

Regulated Specified Components

To maintain the UL listing and compliance to other regulatory certifications and/or declarations, the following regulated components must be used and conditions adhered to. Interchanging or use of other component will void the UL listing and other product certifications and approvals.

Updated product information for configurations can be found on the site at the following URL:

www.QuantaQCT.com

If you do not have access to the Web address, please contact your local representative.

- **Add-in cards:** must have a printed wiring board flammability rating of minimum UL94V-1. Add-in cards containing external power connectors and/or lithium batteries must be UL recognized or UL listed. Any add-in card containing modem telecommunication circuitry must be UL listed. In addition, the modem must have the appropriate telecommunications, safety, and EMC approvals for the region in which it is sold.
- **Peripheral Storage Devices:** must be UL recognized or UL listed accessory and TUV or VDE licensed. Maximum power rating of any one device is 19 watts. Total server configuration is not to exceed the maximum loading conditions of the power supply.

Restriction of Hazardous Substances (RoHS) Compliance

Quanta® Computer Inc. has a system in place to restrict the use of banned substances in accordance with the European Directive 2011/65/EU. Compliance is based on declaration that materials banned in the RoHS Directive are either (1) below all applicable threshold limits or (2) an approved / pending RoHS exemption applies.

RoHS implementation details are not fully defined and may change.

Threshold limits and banned substances are noted below:

- Quantity limit of 0.1% by mass (1000 PPM) for:
 - Lead
 - Mercury
 - Hexavalent Chromium
 - Polybrominated Diphenyl Ethers (PBDE)
- Quantity limit of 0.01% by mass (100 PPM) for:
 - Cadmium

End of Life / Product Recycling

Product recycling and end-of-life take-back systems and requirements vary by country. Contact the retailer or distributor of this product for information about product recycling and / or take-back.